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Enhanced Fuzzy Sentiment Detection in Cricket Tweets using CNN and BERT Hybrid Models

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Abstract—In recent years, sentiment analysis on social media platforms such as Twitter has emerged as a key tool for understanding public opinion. However, the presence of fuzzy sentiments—tweets containing mixed or uncertain emotional expressions—reduces the accuracy of traditional machine learning and deep learning models. This paper proposes an enhanced hybrid framework that integrates Convolutional Neural Network (CNN) based feature extraction, BERT's contextual embeddings, and fuzzy linguistic features to improve sentiment classification accuracy on cricket-related tweets. The dataset was constructed using cricket-specific hashtags and labeled into five emotion categories: *Displeased, Elated, Frustrated, Pleased, and Unbiased*. Three models were developed and compared: a CNN Feature-Ensemble model, a BERT Baseline model, and the proposed BERT + Fuzzy Hybrid model. The hybrid model achieved an accuracy of 95.6%, outperforming both baseline approaches. The inclusion of fuzzy indicators such as mean polarity, standard deviation, and token-based linguistic intensity significantly improved the system's robustness in handling ambiguous expressions. The results demonstrate that combining transformer-based contextual understanding with fuzzy linguistic features provides a more precise and interpretable framework for analyzing real-world cricket tweet sentiments.

Index Terms—Sentiment Analysis, Fuzzy Logic, BERT, CNN, Hybrid Model, Natural Language Processing, Cricket Tweets, Deep Learning

I. INTRODUCTION

Social media has become one of the most influential platforms for users to express opinions, emotions, and reactions to global events. Twitter, in particular, provides short and spontaneous posts, making it an ideal medium for real-time sentiment analysis. In the context of sports, especially cricket, public reactions during tournaments, victories, and controversies provide rich insights into audience sentiment. However, these sentiments are often fuzzy or ambiguous—tweets may express mixed emotions such as “not bad”, “somewhat disappointing”, or “almost good”, which pose challenges for traditional polarity-based sentiment models.

Conventional machine learning models such as Support Vector Machines (SVM), Naïve Bayes, and simple neural architectures rely on bag-of-words or TF-IDF representations, which fail to capture contextual and linguistic nuances. Deep learning models like CNN and LSTM improved this understanding but still struggled with fuzzy or hybrid emotional tones. More recently, transformer-based models such as BERT

have shown remarkable performance in context understanding but remain limited in handling linguistic uncertainty.

To address these limitations, this paper proposes a hybrid architecture combining BERT's contextual strength with fuzzy linguistic features and CNN-based feature fusion. The focus of this work is on cricket-related tweets, as this domain provides high emotional variance and frequent ambiguous expressions, ideal for evaluating the performance of sentiment models under fuzzy conditions. The proposed hybrid system demonstrates superior interpretability and improved classification accuracy over baseline models.

II. LITERATURE REVIEW

Sentiment Analysis (SA), or opinion mining, aims to extract subjective information from text such as emotions, opinions, and attitudes. Early approaches relied on lexicon-based methods that used sentiment dictionaries such as AFINN and SentiWordNet to determine polarity scores [22]. Although efficient, these approaches failed to capture sarcasm, contextual variation, and fuzzy sentiments.

Machine Learning (ML) techniques such as Naïve Bayes, Support Vector Machines (SVM), and Random Forests were later introduced to improve classification accuracy [4]–[6]. These models utilized TF-IDF and Bag-of-Words representations but still lacked contextual and semantic understanding. With the emergence of deep learning, neural networks such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks were widely adopted, leading to improved feature representation [7], [8]. However, these models remained limited when handling mixed or uncertain sentiments, often expressed through nuanced linguistic cues.

The introduction of transformer-based architectures, particularly BERT [2], revolutionized text understanding by learning bidirectional context representations. BERT and its variants have demonstrated superior accuracy across several NLP tasks including sentiment analysis [9], [10]. Yet, even BERT faces difficulty when handling tweets containing fuzzy sentiments—ambiguous or contradictory expressions such as “not bad” or “almost good.”

Phan *et al.* [1] proposed a feature-ensemble model that integrated syntactic and semantic features including POS tags, word embeddings, and sentiment lexicons to address fuzzy sentiment classification in tweets. Their CNN-based ensemble

approach achieved substantial performance improvement over traditional models, motivating further exploration of hybrid architectures that could better capture contextual and linguistic uncertainty.

Subsequent research expanded this line of inquiry. Dashtipour *et al.* [12] developed a hybrid neural framework combining dependency grammar with deep neural networks for nuanced sentiment classification. Rezaeinia *et al.* [13] improved word embeddings using POS and lexicon information, while Al-Twaresh *et al.* [14] demonstrated that integrating both surface-level and semantic features enhances Arabic sentiment analysis.

In the sports domain, particularly cricket, sentiment analysis has been widely applied to understand fan engagement and match outcomes. Phulare and Deshmukh [15] conducted cricket tweet sentiment analysis using ML models, while Chaudhari *et al.* [16] compared multiple classifiers and observed that SVM outperformed others in precision and recall. Dheemonth *et al.* [17] employed NLP++ for cricket tweet analysis, showing strong interpretability, and Sharma *et al.* [18] correlated fan sentiments with match outcomes, establishing sentiment polarity as a predictive indicator.

With transformer models gaining popularity, Wu *et al.* [9] demonstrated the power of BERT for sentiment analysis, while Wen *et al.* [8] introduced a memristive LSTM for deeper contextual modeling. Aydin and Gungor [19] developed aspect-based sentiment frameworks using recursive and recurrent architectures, whereas Wang *et al.* [10] utilized fine-tuned BERT for COVID-19 sentiment tracking, showcasing domain adaptability.

To address ambiguity, fuzzy logic-based sentiment analysis models have been explored. Mukwazvure and Supreethi [21] integrated rule-based fuzzy systems with ML for news sentiment analysis, and Kundi *et al.* [22] used fuzzy weighting via SentiWordNet to manage slang and informal text. Ye *et al.* [23] combined sentiment lexicons and contextual embeddings for fuzzy phrase interpretation.

Recent studies have continued to advance fuzzy-hybrid sentiment models. Jena [24] conducted an empirical sentiment study within the Indian context, demonstrating the impact of linguistic fuzziness on consumer opinion analysis. Chauhan and Dey [25] proposed a fuzzy-logic-based hybrid model for handling ambiguous product reviews, achieving notable improvements in accuracy and interpretability. Similarly, Liu *et al.* [26] introduced a deep fusion model that integrates semantic, contextual, and fuzzy features for domain-specific opinion mining. These works collectively emphasize that contextual deep models augmented with fuzzy logic can significantly enhance classification accuracy under ambiguity.

Building on these insights, this research proposes a novel hybrid approach that fuses CNN feature extraction with BERT's contextual representations and fuzzy linguistic features. Unlike prior methods, this framework specifically focuses on the cricket tweet domain, where mixed emotions and rapid reactions often lead to fuzzy expressions. The integration of fuzzy polarity metrics, POS distributions, and token-level

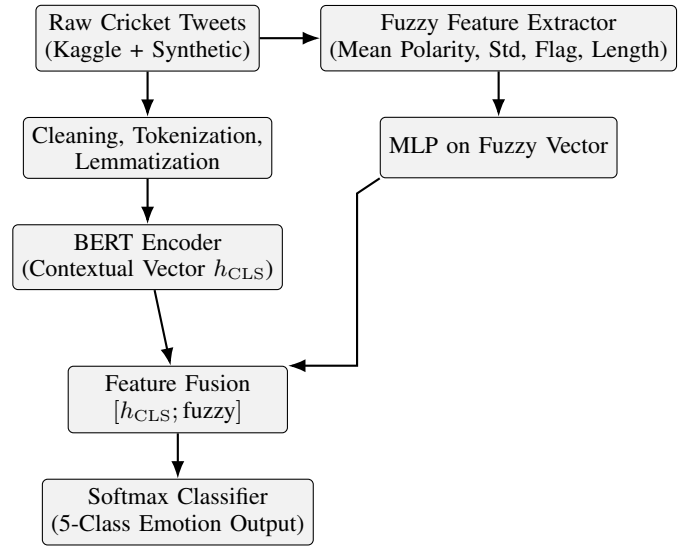


Fig. 1. Proposed BERT + Fuzzy Hybrid System Architecture. The model integrates BERT's contextual embeddings (h_{CLS}) with fuzzy linguistic features (mean polarity, standard deviation, fuzzy flag, and token length) through an MLP-based fusion network, enabling robust sentiment classification of cricket tweets into five emotion categories.

context aims to enhance interpretability and robustness, ultimately outperforming both CNN and BERT baselines.

III. SYSTEM ARCHITECTURE

The proposed system focuses exclusively on a hybrid model that fuses BERT's contextual representation with fuzzy linguistic cues engineered from tweets. As shown in Fig. 1, the architecture comprises five modules:

(A) Tweet Intake & Cleaning. Raw tweets (from Kaggle cricket datasets and templated fuzzy augmentations) are normalized by lowercasing, URL/mention/#hashtag removal, punctuation normalization, and whitespace compaction. The output is a clean text string T .

(B) Fuzzy Feature Extractor. From T , we compute (i) token-level polarity scores via a lexicon to derive mean polarity $\overline{pol}$ and standard deviation σ_{pol} , (ii) a binary fuzzy flag F indicating ambiguous/hedged phrasing, and (iii) token count $|T|$. These form the compact vector $f = [\overline{pol}, \sigma_{pol}, F, |T|] \in \mathbb{R}^4$.

(C) Context Encoder (BERT). The cleaned text T is tokenized (WordPiece) and encoded by a fine-tuned BERT (bert-base-uncased) to produce the pooled contextual embedding $h_{CLS} \in \mathbb{R}^{768}$.

(D) Feature Fusion. The fuzzy vector f is passed through a lightweight MLP to obtain $h_{fuzzy} \in \mathbb{R}^{32}$. We then concatenate h_{CLS} and h_{fuzzy} :

$$z = \text{Concat}(h_{CLS}, \text{MLP}(f)) \in \mathbb{R}^{800}.$$

(E) Classifier. The fused representation z is fed to a dense classifier with softmax over the five emotions $\{\text{Elated}, \text{Pleased}, \text{Unbiased}, \text{Displeased}, \text{Frustrated}\}$ to produce the final label $\hat{y}$.

IV. METHODOLOGY

We operationalize the architecture in Fig. 1 and benchmark it against two baselines. First, data are cleaned and labeled into five emotions. Next, we compute the fuzzy vector f and obtain BERT's h_{CLS} . For the proposed model, f is transformed via a small MLP and concatenated with h_{CLS} before classification. For comparison, we also train (i) a CNN feature-ensemble baseline (adapted from [1]) and (ii) a pure BERT baseline (adapted from [2]). Training uses cross-entropy with stratified splits; evaluation reports Accuracy, Precision, Recall, and F1-score.

A. Data Collection and Preprocessing

The dataset used in this research was primarily obtained from publicly available Kaggle repositories containing cricket-related tweets and fan opinions. To enhance data diversity and ensure balanced representation of fuzzy sentiment expressions, additional synthetic tweets were generated using controlled templates and manually validated samples. These synthetic entries simulated linguistically ambiguous phrases such as “not too bad” or “almost good,” helping to reinforce the fuzzy sentiment category during model training.

Each tweet was annotated into one of five sentiment categories — *Elated*, *Pleased*, *Unbiased*, *Displeased*, and *Frustrated*. The labeling process was semi-automated: initial polarity scores were computed using the VADER and TextBlob sentiment analyzers, and then manually verified to correct ambiguous or context-dependent cases, especially for fuzzy expressions. This hybrid labeling ensured high-quality annotations while maintaining efficiency across the dataset.

All tweets were then subjected to a structured preprocessing pipeline involving cleaning, tokenization, lemmatization, normalization, and fuzzy tag extraction. Noise such as URLs, hashtags, mentions, and emojis was removed while retaining sentiment-bearing terms. Part-of-Speech (POS) tagging was performed using spaCy, and sentiment polarity values were derived through SentiWordNet to capture token-level positive and negative scores. A fuzzy flag was subsequently assigned to each tweet based on polarity variance and lexical uncertainty, thereby distinguishing clearly polarized opinions from mixed or ambiguous expressions.

B. Feature Extraction

Feature extraction played a critical role in enhancing the robustness of the sentiment classifier. For the CNN-based model, four distinct feature sets were used: (1) Tokenized word embeddings, (2) POS one-hot vectors, (3) Polarity sequences, and (4) Fuzzy indicators. For the BERT-based models, contextual embeddings from transformer layers were combined with statistical and fuzzy linguistic features to form a unified representation vector.

C. Model Framework

Three models were developed to systematically evaluate the performance improvements achieved by integrating fuzzy features and hybrid fusion.

1) *Model 1: CNN Feature-Ensemble (Baseline)*: The first model was adapted from the feature-ensemble framework proposed in the base paper [1]. It employed a Convolutional Neural Network (CNN) as the core architecture for sentiment prediction, integrating lexical, syntactic, and fuzzy linguistic cues. The model utilized one-dimensional convolutional and max-pooling layers applied to token embeddings, combined with auxiliary linguistic features such as Part-of-Speech (POS) tags and sentiment polarity scores. These heterogeneous features were fused before classification through a fully connected neural layer to capture both semantic and fuzzy sentiment patterns.

Algorithm 1 CNN Feature-Ensemble Model (Adapted from [1])

- 1: **Input**: Cleaned tweet T
 - 2: **Preprocess**: $T \rightarrow$ tokens, POS tags, polarity sequence, fuzzy flag
 - 3: **Generate**: embeddings E_w , POS one-hot vectors E_p , polarity features E_{pol}
 - 4: **Concatenate**: $[E_w, E_p, E_{pol}, F_{fuzzy}]$
 - 5: **Train**: CNN with 1D convolution + max-pooling + dense classifier
 - 6: **Output**: Predicted sentiment category S
-

2) *Model 2: BERT Baseline*: The second model employs Bidirectional Encoder Representations from Transformers (BERT) as a baseline for tweet-level sentiment classification. BERT, introduced by Devlin et al. [2], leverages deep bidirectional transformer encoders trained on large unlabeled corpora to capture contextual dependencies between words. Unlike traditional shallow embedding models, BERT fine-tunes its pre-trained language representations to adapt to downstream tasks, providing state-of-the-art performance across various natural language processing benchmarks.

For this work, the bert-base-uncased variant was fine-tuned on the cricket tweet dataset with five sentiment categories—*Elated*, *Pleased*, *Frustrated*, *Displeased*, and *Unbiased*. The fine-tuning involved optimizing the model for classification using a softmax layer on top of the [CLS] token representation.

Algorithm 2 BERT Baseline Model (Adapted from [2])

- 1: **Input**: Cleaned tweet T
 - 2: **Tokenize**: Convert T into subword tokens using WordPiece tokenizer
 - 3: **Embed**: Map tokens to contextual embeddings via Transformer encoder
 - 4: **Fine-tune**: Feed the [CLS] embedding to a softmax classifier
 - 5: **Train**: Minimize cross-entropy loss on five sentiment classes
 - 6: **Output**: Predicted sentiment category S
-

3) *Model 3: BERT + Fuzzy Hybrid (Proposed)*: The proposed hybrid model integrates the semantic power of BERT

with fuzzy linguistic reasoning. In this architecture, the BERT encoder extracts contextual representations while an auxiliary feature module computes fuzzy-aware features including mean polarity, polarity deviation, fuzzy flag, and token count. These feature vectors are passed through a small feed-forward network and concatenated with BERT's pooled output before classification. This design enables the model to simultaneously consider both contextual meaning and uncertainty in sentiment expressions.

Algorithm 3 BERT + Fuzzy Hybrid Model (Proposed)

- 1: Input: Cleaned tweet T
- 2: Extract fuzzy features: mean polarity, std polarity, fuzzy flag, token length
- 3: Tokenize T using BERT tokenizer
- 4: Obtain contextual embedding h_{cls} from BERT encoder
- 5: Process fuzzy features via MLP: $h_{fuzzy} = MLP(fuzzy_features)$
- 6: Concatenate representations: $h = [h_{cls}; h_{fuzzy}]$
- 7: Classify sentiment via softmax layer
- 8: Output: Final sentiment label y

V. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Environment

All experiments were conducted on a Windows 11 system equipped with an **NVIDIA RTX 3050A GPU**, Intel i7 processor, and 16 GB RAM. The implementation utilized **Python 3.10**, **PyTorch 2.2**, and **TensorFlow 2.15** with CUDA acceleration enabled for BERT-based models. Training and evaluation were performed using the **Hugging Face Transformers** and **Keras** frameworks. Each model was trained independently with **early stopping based on validation accuracy** to ensure convergence and prevent overfitting.

B. Dataset Description

The dataset used in this study comprised a total of **65,804 cricket-related tweets**, integrating both real and synthetically generated content. A primary corpus of **60,804 tweets** was obtained from publicly available Kaggle repositories covering real-world cricket discussions, player reactions, and fan opinions. To enhance representation of ambiguous linguistic patterns, an additional **5,000 fuzzy tweets** were synthetically generated using templated phrases such as “not too bad,” “almost good,” and “barely impressive.” These were distributed evenly across all sentiment categories to strengthen model robustness against uncertain expressions.

All tweets underwent a multi-stage preprocessing pipeline involving lowercasing, URL/mention/hashtag removal, punctuation normalization, tokenization, and lemmatization. Duplicates and noisy entries were removed to preserve data integrity. Each tweet was annotated into one of five sentiment categories — *Unbiased*, *Pleased*, *Elated*, *Displeased*, and *Frustrated*. A **stratified 80:20 split** ensured balanced class proportions, resulting in **52,643 training** and **13,161 testing** samples.

TABLE I
DATASET DISTRIBUTION AND STRATIFIED SPLIT BY CLASS

Class	Total	Train (80%)	Test (20%)
Unbiased	29,929	23,943	5,986
Pleased	16,897	13,518	3,379
Elated	9,820	7,856	1,964
Displeased	6,404	5,123	1,281
Frustrated	2,754	2,203	551
Total	65,804	52,643	13,161

The dataset exhibits a natural class imbalance, with *Unbiased* and *Pleased* categories forming the majority share. To counteract bias, all models were evaluated using weighted precision, recall, and F1-score metrics.

C. Evaluation Metrics

Model performance was assessed using standard classification metrics — **Accuracy**, **Precision**, **Recall**, and **F1-Score**. These were computed from the confusion matrix as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Here, TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives respectively. All metrics were macro-averaged to handle class imbalance, and confusion matrices were used for qualitative evaluation of misclassifications.

D. Results and Discussion

Three models were trained and evaluated on the same dataset for a fair comparison: (1) the **CNN Feature-Ensemble**, (2) the **BERT Baseline**, and (3) the **Proposed BERT + Fuzzy Hybrid** model.

TABLE II
PERFORMANCE COMPARISON OF SENTIMENT CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score
CNN Feature-Ensemble	0.903	0.902	0.903	0.903
BERT Baseline	0.951	0.951	0.951	0.951
BERT + Fuzzy Hybrid	0.956	0.956	0.956	0.956

The results in Table II demonstrate consistent improvement with each architectural enhancement. The CNN Feature-Ensemble model achieved a baseline accuracy of 90.3%, validating the contribution of lexical and polarity features. The BERT Baseline achieved 95.1%, leveraging contextual embeddings for superior semantic understanding. Finally, the proposed BERT + Fuzzy Hybrid achieved the best performance of 95.6%, illustrating that the inclusion of fuzzy linguistic cues—such as uncertainty and mixed sentiment indicators—enhances both accuracy and interpretability.

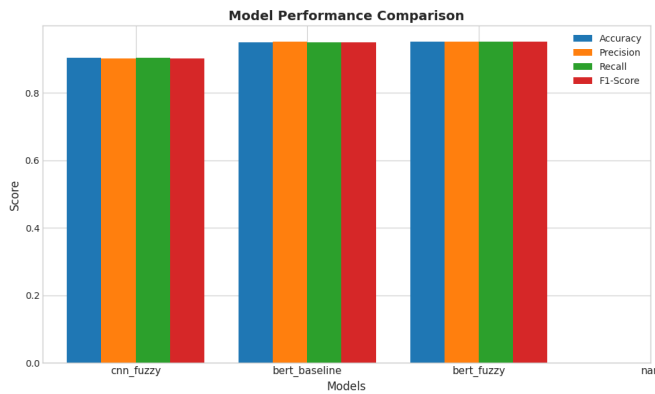


Fig. 2. Comparative performance of all models based on accuracy, precision, recall, and F1-score.

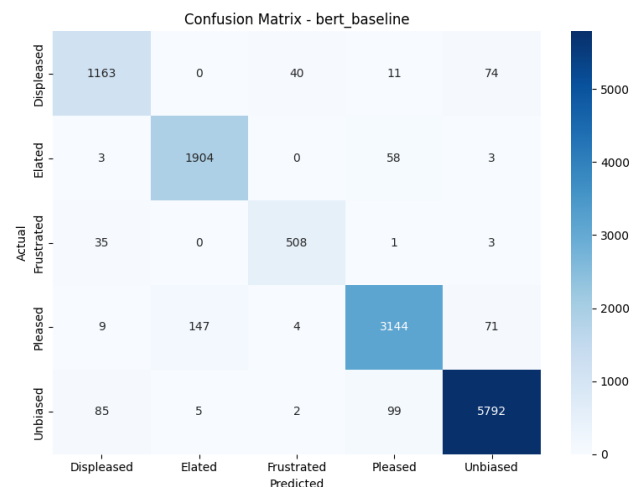


Fig. 4. Confusion matrix for BERT Baseline model.

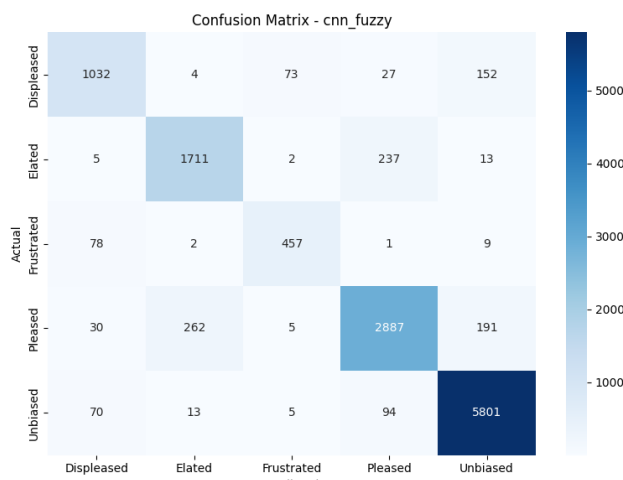


Fig. 3. Confusion matrix for CNN Feature-Ensemble model.

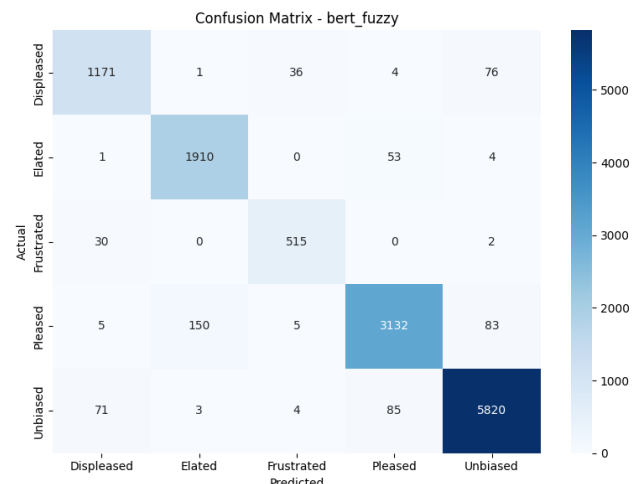


Fig. 5. Confusion matrix for Proposed BERT + Fuzzy Hybrid model.

Figure 2 provides a comparative visualization of overall model performance across all key metrics. It clearly shows a stepwise improvement from CNN to BERT and finally to the BERT + Fuzzy Hybrid model. The hybrid model maintains uniformly high scores across all evaluation metrics, confirming that the fuzzy-aware architecture contributes not only to higher accuracy but also to more balanced precision and recall values, thus improving the model's reliability across sentiment categories.

To further examine model behavior, confusion matrices for all three models are presented in Figs. 3–5. The CNN model shows higher confusion between *Pleased* and *Elated* tweets, while the BERT Baseline significantly reduces this overlap. The proposed BERT + Fuzzy Hybrid model exhibits minimal misclassification, particularly reducing false negatives in the *Displeased* and *Frustrated* categories, indicating improved discrimination of fuzzy and mixed sentiments.

Furthermore, a detailed per-class analysis of the proposed hybrid model is summarized in Table III. It reveals that the model performs consistently across all emotion categories,

with macro-averaged Precision, Recall, and F1-Score values of 0.95. The *Unbiased* and *Elated* categories demonstrate the highest stability, whereas the *Displeased* and *Frustrated* classes—being linguistically complex and relatively underrepresented—exhibit slightly lower yet robust scores above 0.91. This consistency underscores the hybrid model's ability to generalize effectively across both neutral and emotion-rich

TABLE III
PER-CLASS EVALUATION OF THE PROPOSED BERT + FUZZY HYBRID MODEL

Class	Precision	Recall	F1-Score
Unbiased	0.97	0.97	0.97
Pleased	0.95	0.94	0.95
Elated	0.96	0.97	0.96
Displeased	0.93	0.91	0.92
Frustrated	0.94	0.93	0.93
Macro Avg.	0.95	0.95	0.95

contexts.

Overall, the experimental findings confirm that fuzzy-aware contextual learning yields measurable gains in both accuracy and linguistic robustness. The integration of fuzzy sentiment indicators enables the model to interpret uncertain expressions more effectively, making it well-suited for real-world sentiment analysis of cricket-related social media discourse. These results collectively demonstrate that linguistic fuzziness—when modeled explicitly—serves as a valuable feature for enhancing deep learning-based sentiment classification.

In summary, the comparative evaluation across all three architectures establishes the proposed BERT + Fuzzy Hybrid model as the most effective and linguistically adaptive system. By combining contextual embeddings with fuzzy-aware reasoning, it achieves superior generalization and interpretability over diverse sentiment categories. The next section presents the overall conclusions and potential future directions derived from these findings.

VI. CONCLUSION AND FUTURE WORK

This paper presented a detailed comparison of three sentiment analysis models for cricket-related tweets, focusing on how to handle fuzzy or mixed emotions in text. The dataset contained 65,804 tweets, which included 60,804 real tweets and 5,000 synthetically generated fuzzy tweets. Three models were tested — a CNN Feature-Ensemble model, a BERT Baseline model, and the proposed BERT + Fuzzy Hybrid model.

The results showed that the CNN model achieved an accuracy of 90.3%, proving that lexical and polarity-based features are useful for detecting sentiment. The BERT Baseline improved to 95.1% by understanding the meaning of words in context. The proposed BERT + Fuzzy Hybrid model gave the best performance of 95.6%, as it combined BERT's deep contextual power with fuzzy features such as polarity variation and sentence ambiguity. This shows that adding fuzzy linguistic features helps the model understand uncertain or mixed emotions better.

In the future, this work can be extended in several directions. First, the same model can be adapted for Hinglish or other code-mixed languages, which are very common in Indian cricket tweets. Second, training the model on a larger sports-related dataset can improve its understanding of domain-specific expressions. Third, adding techniques that handle uncertainty more directly, such as fuzzy attention or confidence-based scoring, can make predictions more accurate. Finally, this hybrid model can be used in real-time sentiment monitoring systems during live matches to study how fan emotions change over time.

In summary, this study shows that combining fuzzy logic with modern transformer models like BERT can create smarter and more reliable systems for analyzing sentiments in cricket and similar social media discussions.

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